Fructose is more soluble than other sugars and hard to crystallize because it is more hygroscopic and holds onto water stronger than the others. This means that fructose can be used to extend the shelf life of baked products more than other sugars.

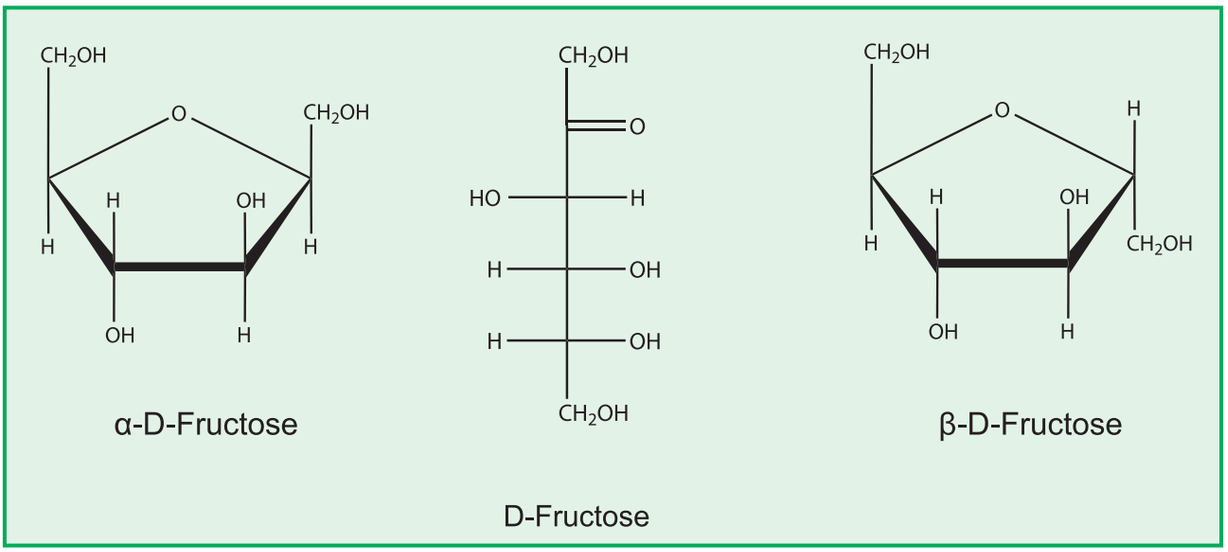


Figure 5.4 Structures of Fructose

5.3.3 Galactose

Galactose is a monosaccharide and has the same chemical formula as glucose, i.e., $C_6H_{12}O_6$. It is similar to glucose in its structure, differing only in the position of one hydroxyl group. This difference, however, gives galactose different chemical and biochemical properties to glucose. In solution, it forms 5- and 6-membered rings but also exists in linear form. Small amounts of lactose and galactose can appear in nondairy foods.

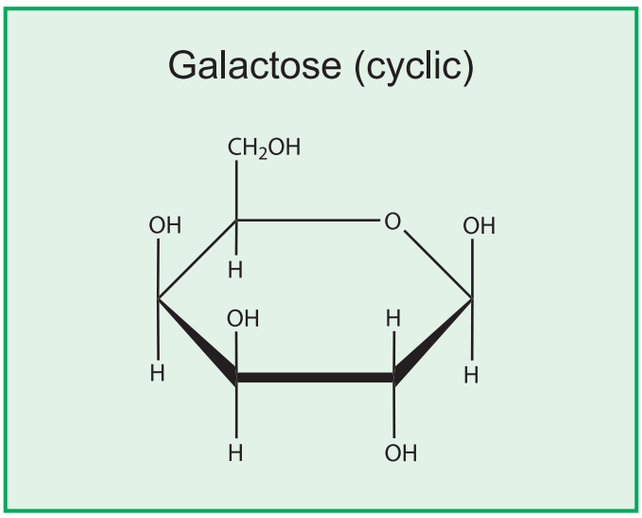


Figure 5.5 Pyranose structure of galactose

Unit 5.indd 127 21-05-2018 15:10:27